Improved single differential cross-section results

Chatura Kuruppu

New Mexico State University

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Outline

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2. Event Selection Used (DocDB 2111-V42 & Fiducial Cuts)
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7. Acceptance Corrections (LH2 & LD2)
8. Cross-Section Comparison All Roadsets (LH2 & LD2)

Events from runs 2 and 3 (Input Files Used)

Note: Currently using all the runs saved in runs 2 and 3.

Harsha's ROOT files:
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/:

- merged_RS57_LH2_1_1138.root
- merged_RS57_LD2_3_1138.root
- merged_RS57_Empty_2_1138.root
- merged_RS59_LH2_1_465.root
- merged_RS59_LD2_3_466.root
- merged_RS59_Empty_2_466.root
- merged_RS62_LH2_1_1234.root
- merged_RS62_LD2_3_1237.root
- merged_RS62_Empty_2_1234.root
- merged_RS70_LH2_1_264.root
- merged_RS70_LD2_3_266.root
- merged_RS70_Empty_2_267.root

Abi's ROOT files:
/seaquest/users/apun/e906_projects/rs67_merged_files/:

- merged_RS67_3089LH2.root
- merged_RS67_3089LD2.root
- merged_RS67_3089flask.root

Event Selection Used: Chuck Cuts (DocDB 2111-V42) & Fiducial Cuts

Chuck Cuts (DocDB2111-V42**)** with beam_offset $= 1.6$ cm and mass cut $= 4.2$ GeV/c^2:

Dimuon Cuts (dimuon_cut_2111_v42)

- Vertex: $|dx| < 0.25$, $|dy - 1.6| < 0.22$, $-280 < dz < -5$ cm
- Transverse Vertex: $dx^2 + (dy - 1.6)^2 < 0.06$ cm^2
- Momentum: $|dp_x| < 1.8$, $|dp_y| < 2.0$, $dp_x^2 + dp_y^2 < 5.0$ (GeV/c)^2, $38 < dp_z < 116$ GeV/c
- Kinematics: $4.2 <$ mass < 8.8 GeV/c^2, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, $|\cos\theta| < 0.5$
- Fit & Separation: $|$trackSeparation$| < 270$ cm, $\chi^2_{\text{dimuon}} < 18$

Single Track Cuts (track1_cut & track2_cut, $i = 1, 2$)

- Target/Dump χ^2: $\chi^2_{i,\text{target}} < 15$, $\chi^2_{i,\text{target}} < 1.5\chi^2_{i,\text{upstream}}$, $\chi^2_{i,\text{target}} < 1.5\chi^2_{i,\text{dump}}$
- Track Quality: nHits$_i > 13$, $\chi^2_i/(\text{nHits}_i - 5) < 12$, $-320 < z_{i,v} < -5$ cm
- Apertures: $x^2_{i,t} + (y_{i,t} - 1.6)^2 < 320$, $16 < x^2_{i,d} + (y_{i,d} - 1.6)^2 < 1100$ cm^2
- Station 1 & 3: $9 < p^{\text{st1}}_{z,i} < 75$, $|p^{\text{st1}}_{y,i}| > 0.02$ GeV/c, $y_{i,\text{st1}}/y_{i,\text{st3}} < 1$, $y_{i,\text{st1}} \cdot y_{i,\text{st3}} > 0$
- Δp Matching: $||p^{\text{st1}}_{x,i} - p^{\text{st3}}_{x,i}| - 0.416| < 0.008$, $|p^{\text{st1}}_{y,i} - p^{\text{st3}}_{y,i}| < 0.008$,

Track Pair Cuts (tracks_cut_2111_v42)

- $|\chi^2_{1,\text{target}} + \chi^2_{2,\text{target}} - \chi^2_{\text{dimuon}}| < 2.0$
- $y_{1,\text{st3}} \cdot y_{2,\text{st3}} < 0$ (opposite top/bottom halves, no beam offset)
- nHits$_1$ + nHits$_2 > 29$ and nHits1St1 + nHits2St1 > 8
- $|x_{1,\text{st1}} + x_{2,\text{st1}}| < 42$ cm

Occupancy Cuts (occ_cut_2111_v42)

- $D_1 < 400$, $D_2 < 400$, $D_3 < 400$, and $D_1 + D_2 + D_3 < 1000$
- ***Note:*** **Intensity cuts (**intensity_cut_2111_v42**) are not applied.**

Additional Fiducial Cut Applied

In addition to the Chuck cuts above, we apply the kinematic fiducial cut:

$(4.2 <$ mass < 8.8 GeV/c^2) & ($0.0 < x_F < 0.8$) & ($0 < p_T < 1.8$ GeV/c)

Changes made

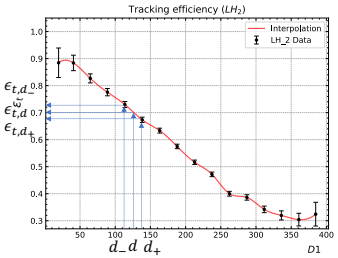
- Recalculated acceptance correction for each roadset using Kenichi's new MC files
- Recalculated reconstruction efficiencies by using efficiency curves provided by Kenichi for each roadset for both LH2 and LD2 targets

$$\frac{\mathrm{d}\sigma_{pp}}{\mathrm{d}p_T} = \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta p_T}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right] \tag{1}$$

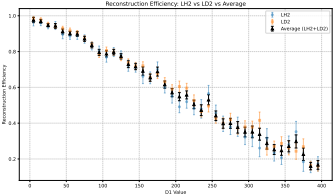
$$\begin{aligned}\frac{\mathrm{d}\sigma_{pd}}{\mathrm{d}p_T} = \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta p_T}\Bigg(&\left[\frac{Y^{\mathrm{LD2}}_{\mathrm{total}} - Y^{\mathrm{LD2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LD2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LD2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right] \\ &- \frac{T_{HD}}{T_{HH}}\frac{I_{\mathrm{LD2}}}{I_{\mathrm{LH2}}}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right]\Bigg)\end{aligned} \tag{2}$$

New Reconstruction Efficiency Correction (D1-10)

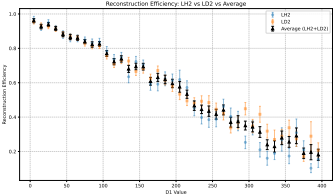
Previous Global Curve (All RS)



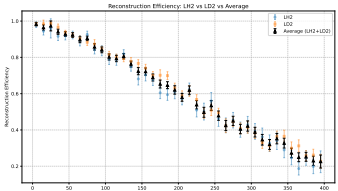
RS57 Efficiency Curve



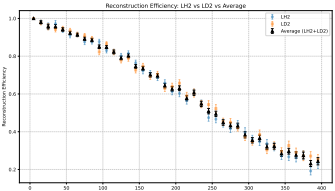
RS59 Efficiency Curve



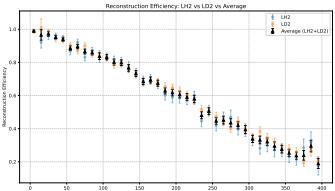
RS62 Efficiency Curve



RS67 Efficiency Curve

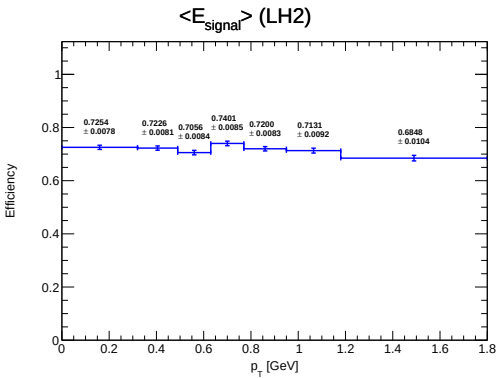


RS70 Efficiency Curve

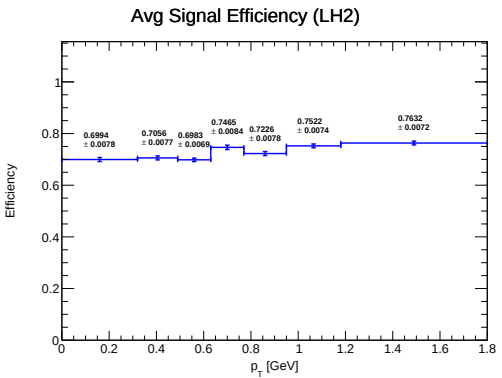


RS57: Latest Vs Existing Reconstruction Efficiencies (LH2)

Latest Efficiency (RS57 LH2)

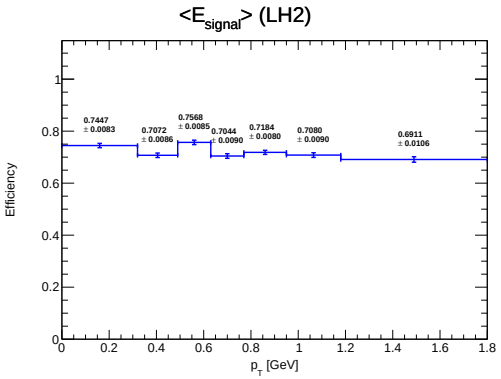


Existing Efficiency (RS57 LH2)

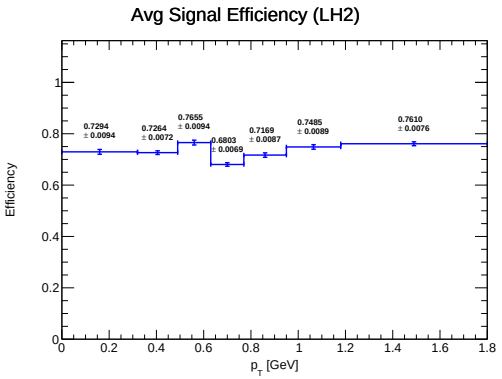


RS59: Latest Vs Existing Reconstruction Efficiencies (LH2)

Latest Efficiency (RS59 LH2)

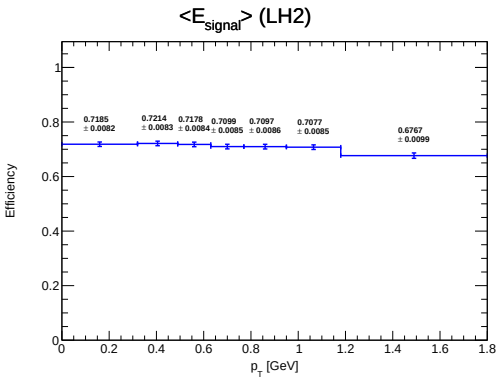


Existing Efficiency (RS59 LH2)

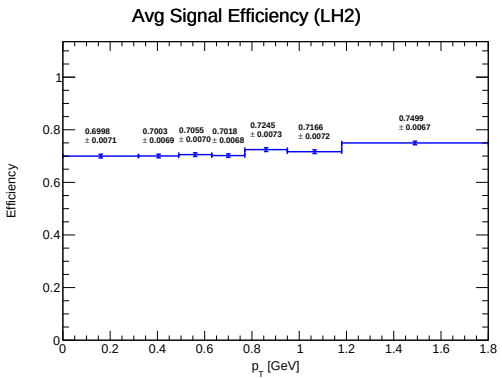


RS62: Latest Vs Existing Reconstruction Efficiencies (LH2)

Latest Efficiency (RS62 LH2)

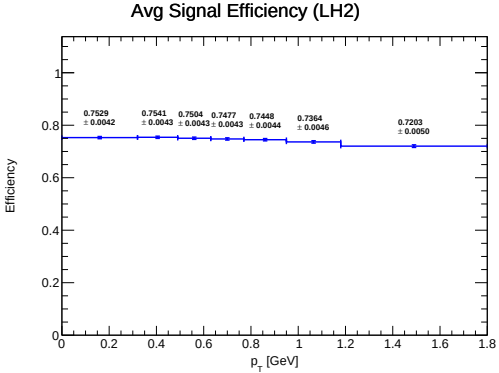


Existing Efficiency (RS62 LH2)

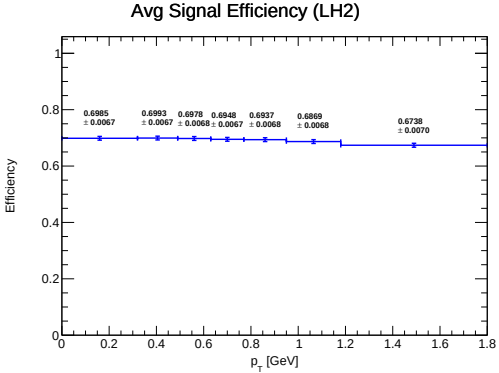


RS67: Latest Vs Existing Reconstruction Efficiencies (LH2)

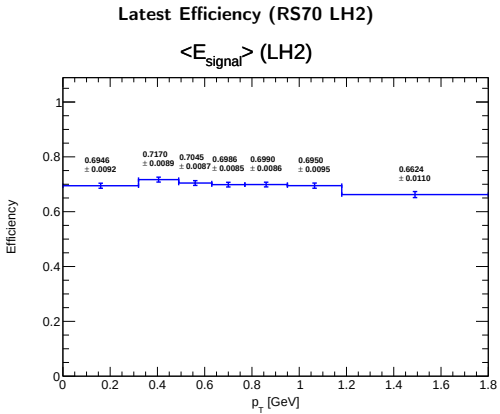
Latest Efficiency (RS67 LH2)

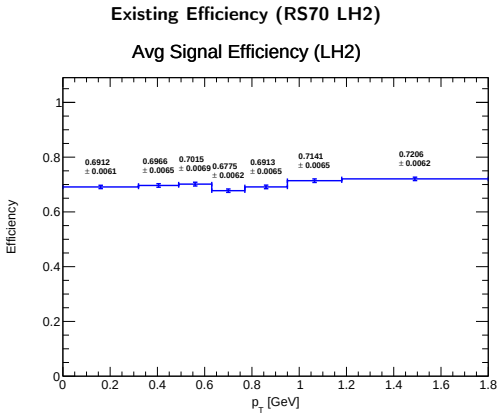


Existing Efficiency (RS67 LH2)



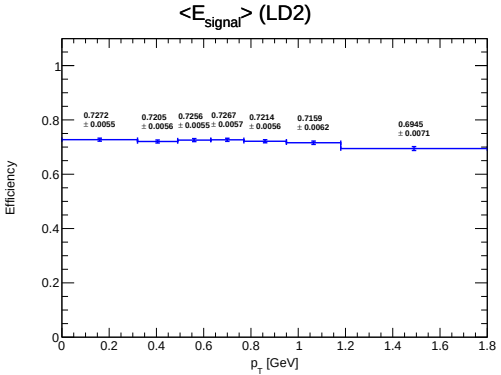
RS70: Latest Vs Existing Reconstruction Efficiencies (LH2)



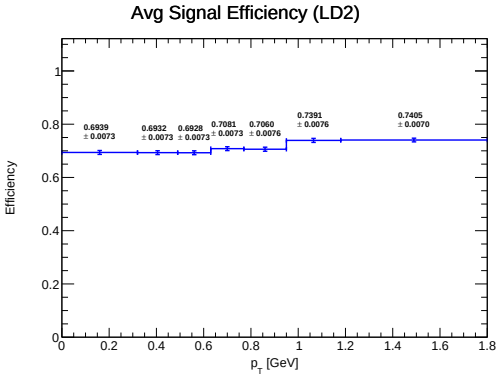


RS57: Latest Vs Existing Reconstruction Efficiencies (LD2)

Latest Efficiency (RS57 LD2)

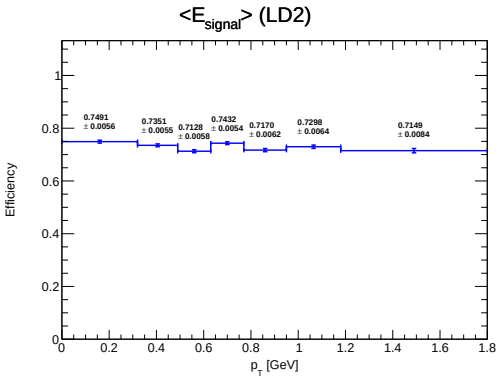


Existing Efficiency (RS57 LD2)

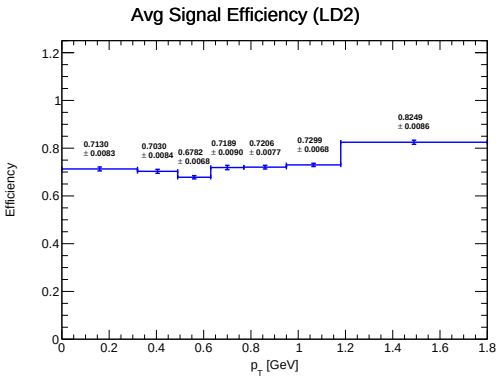


RS59: Latest Vs Existing Reconstruction Efficiencies (LD2)

Latest Efficiency (RS59 LD2)

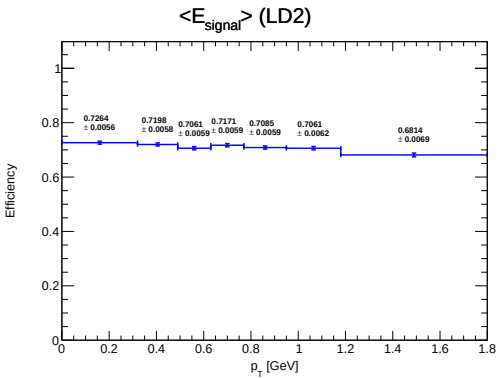


Existing Efficiency (RS59 LD2)

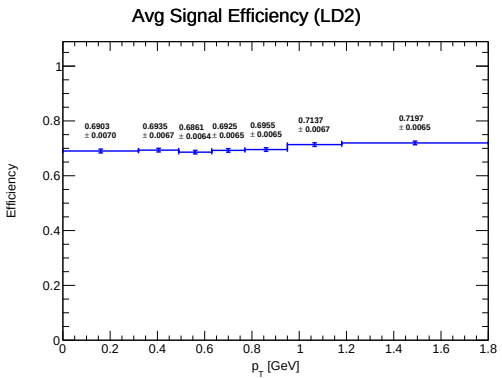


RS62: Latest Vs Existing Reconstruction Efficiencies (LD2)

Latest Efficiency (RS62 LD2)

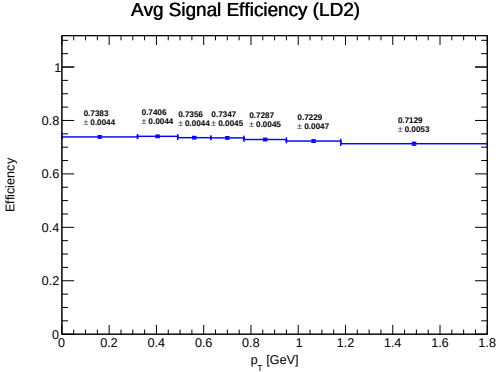


Existing Efficiency (RS62 LD2)

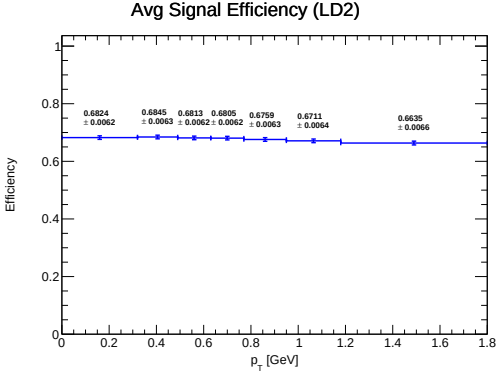


RS67: Latest Vs Existing Reconstruction Efficiencies (LD2)

Latest Efficiency (RS67 LD2)

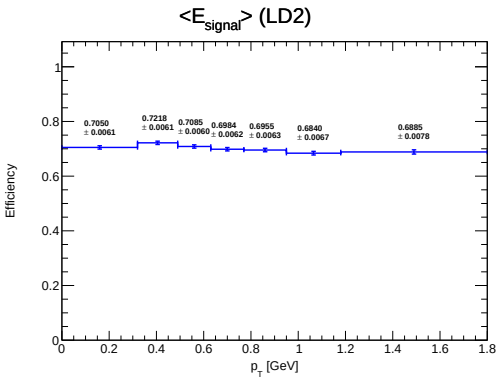


Existing Efficiency (RS67 LD2)

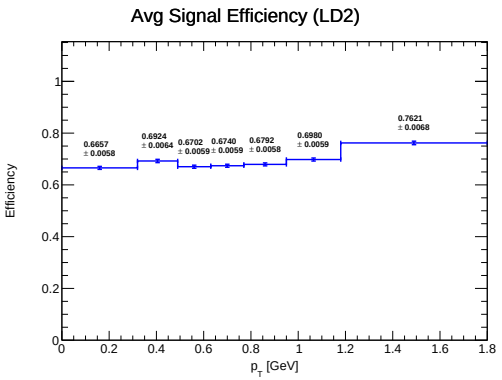


RS70: Latest Vs Existing Reconstruction Efficiencies (LD2)

Latest Efficiency (RS70 LD2)

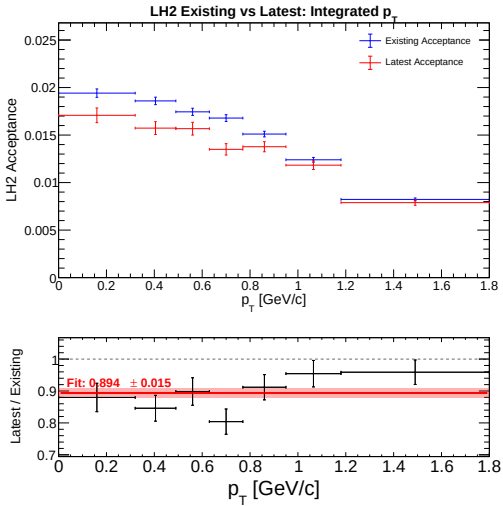


Existing Efficiency (RS70 LD2)

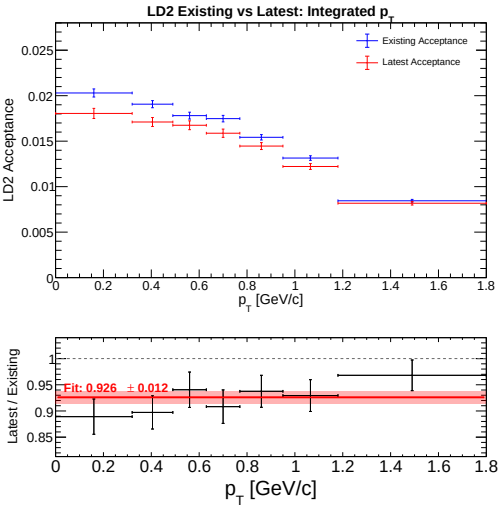


RS57 Acceptance Corrections

LH2 (RS57)

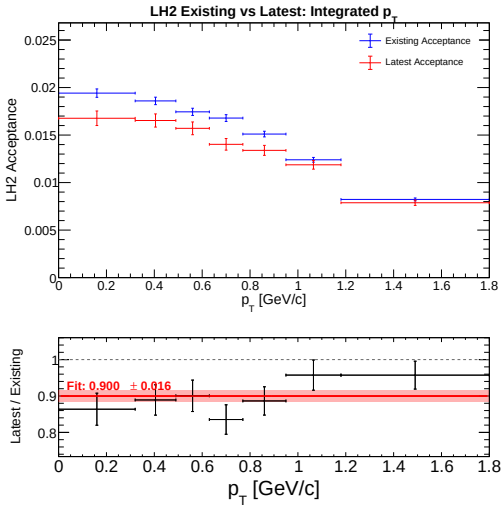


LD2 (RS57)

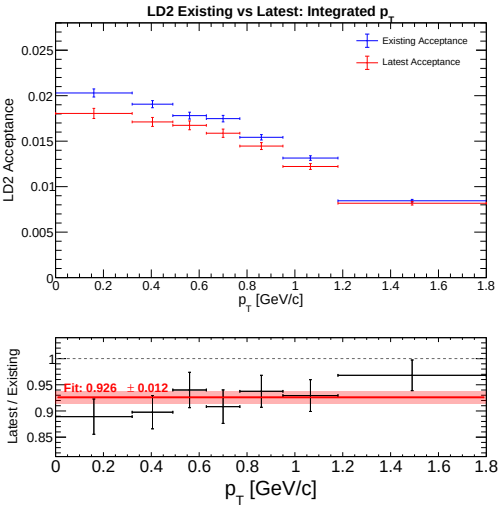


RS59 Acceptance Corrections

LH2 (RS59)

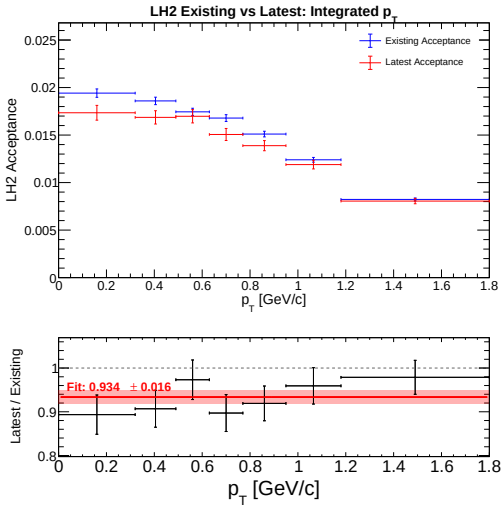


LD2 (RS59)

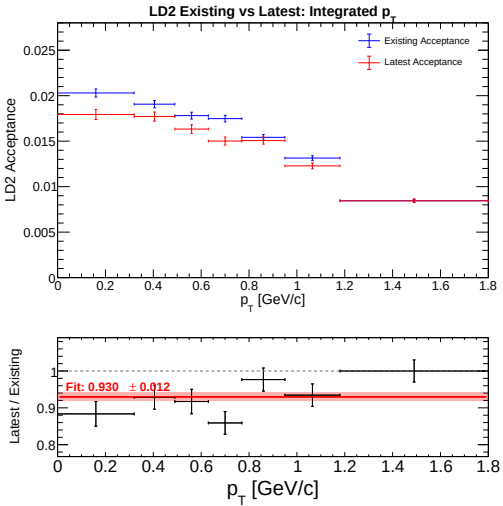


RS62 Acceptance Corrections

LH2 (RS62)

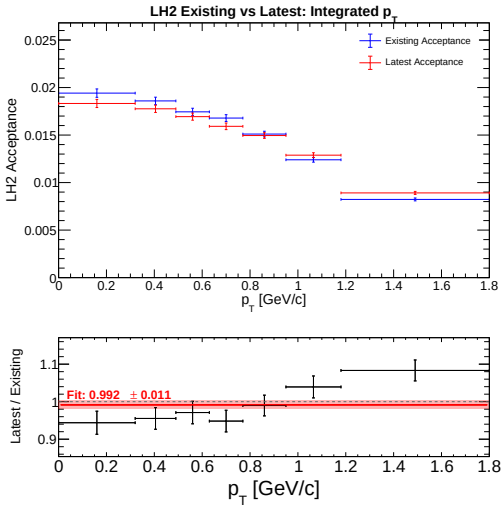


LD2 (RS62)

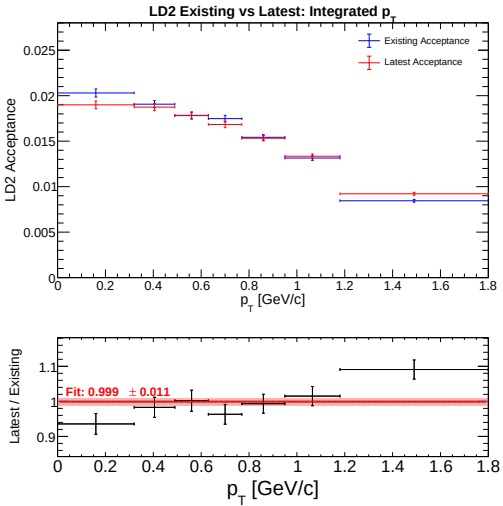


RS67 Acceptance Corrections

LH2 (RS67)

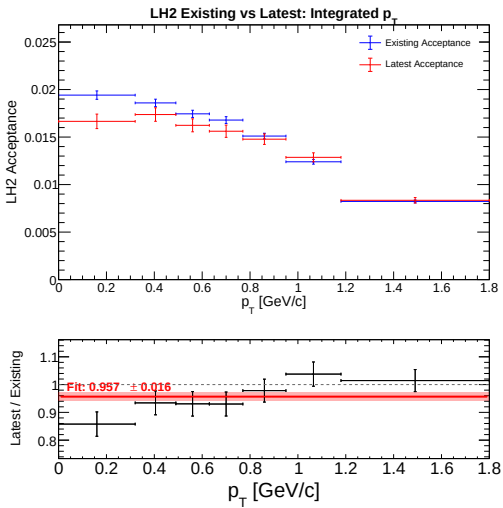


LD2 (RS67)

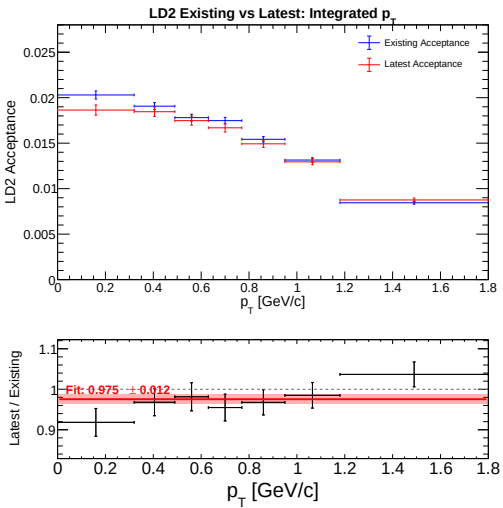


RS70 Acceptance Corrections

LH2 (RS70)

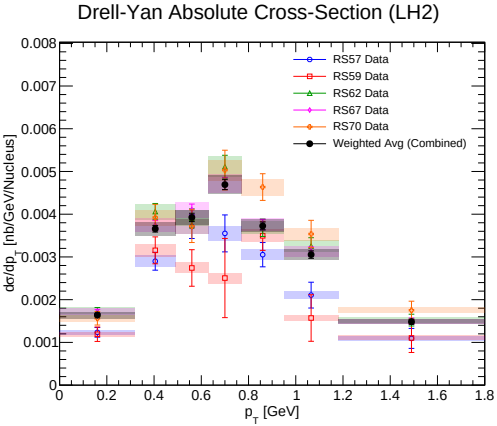


LD2 (RS70)

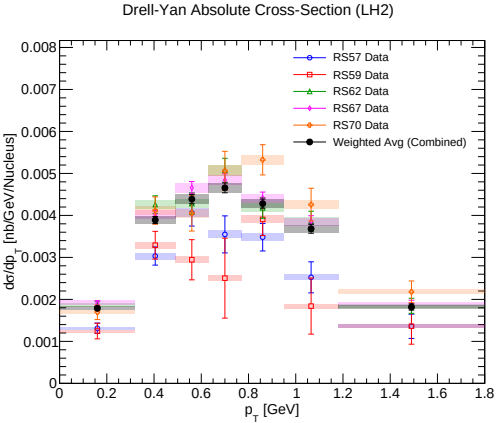


LH2 Cross-Section Comparison All Roadsets (pT bins)

Latest

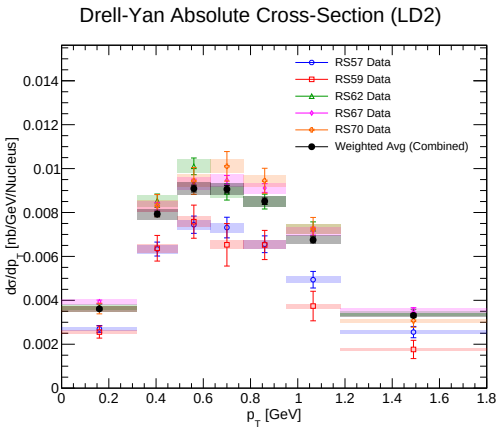


Existing

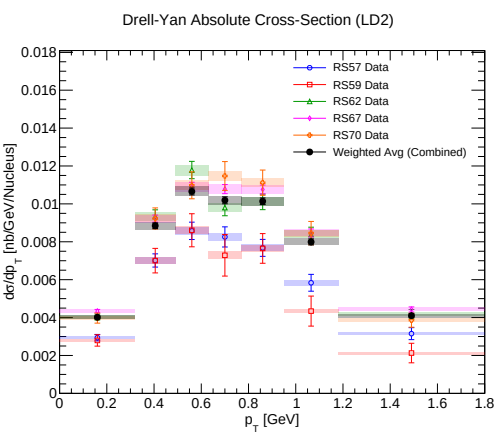


LD2 Cross-Section Comparison All Roadsets (pT bins)

Latest

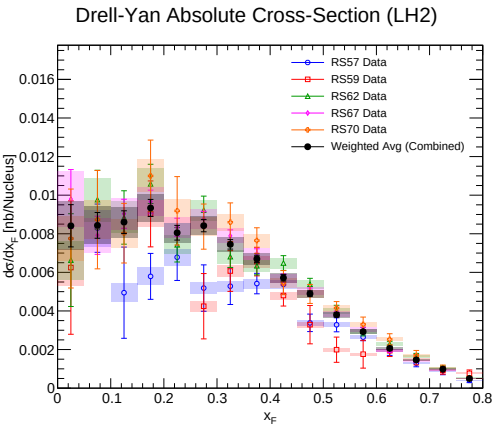


Existing

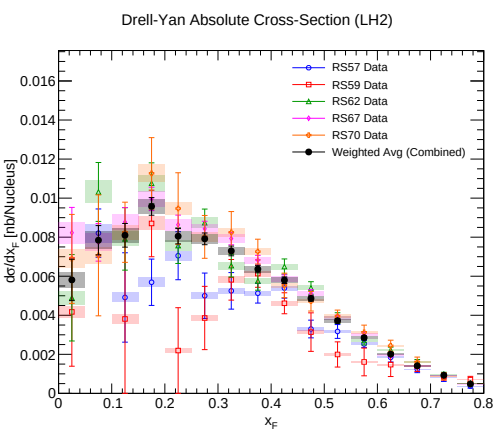


LH2 Cross-Section Comparison All Roadsets (xF bins)

Latest

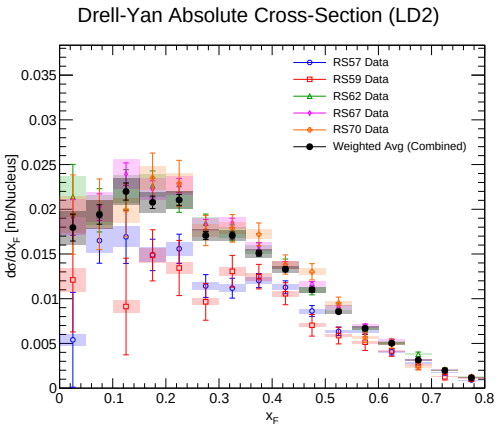


Existing

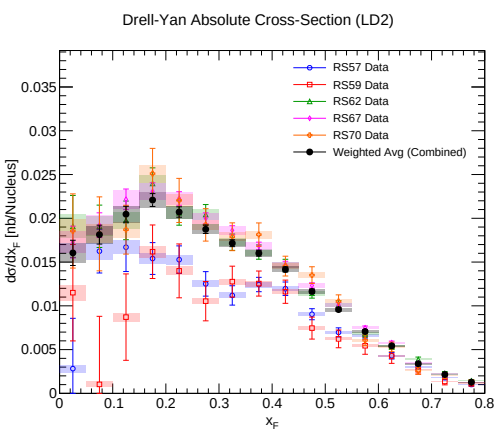


LD2 Cross-Section Comparison All Roadsets (xF bins)

Latest

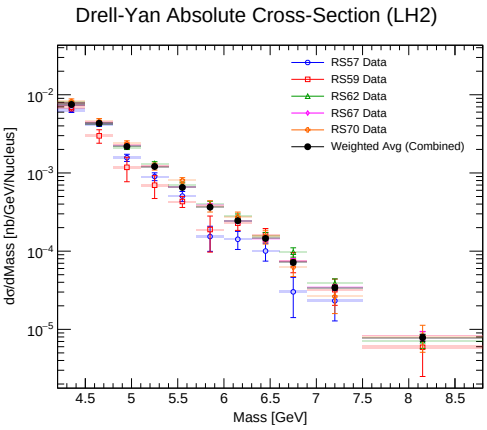


Existing

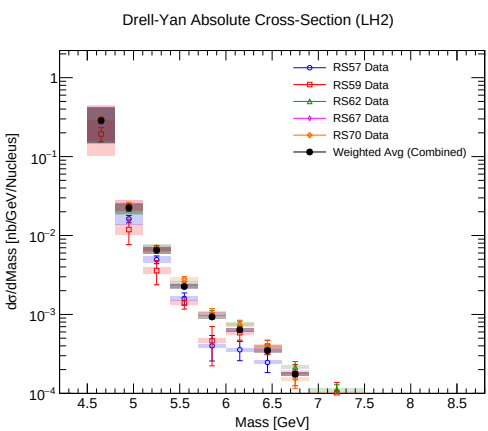


LH2 Cross-Section Comparison All Roadsets (Mass bins)

Latest

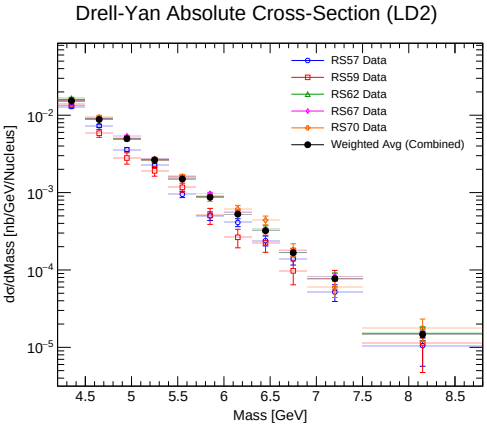


Existing

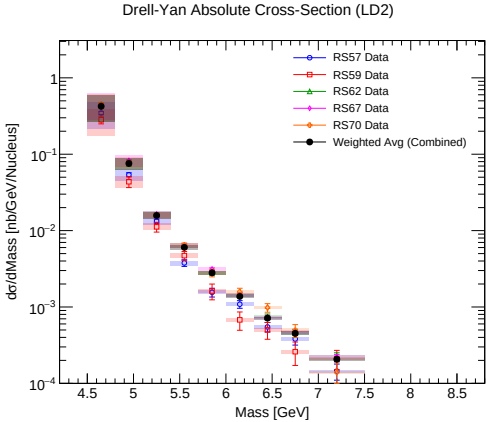


LD2 Cross-Section Comparison All Roadsets (Mass bins)

Latest

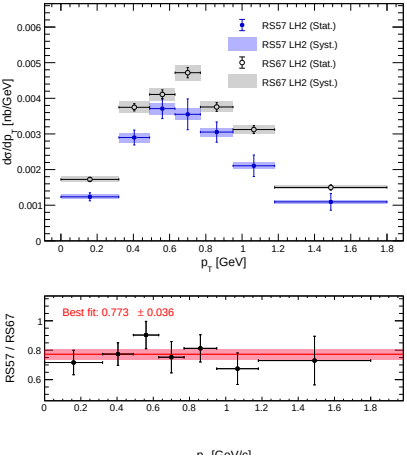


Existing

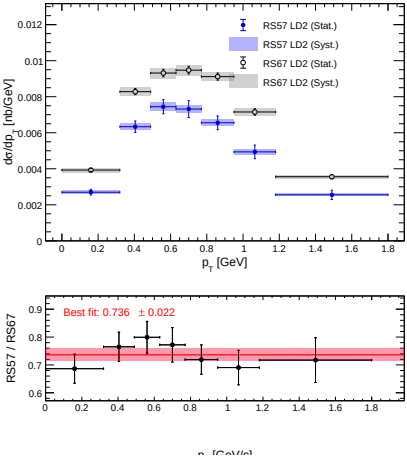


Cross-Section Ratio RS57 to RS67

LH2

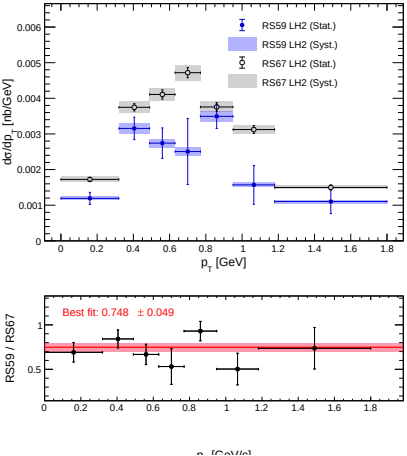


LD2

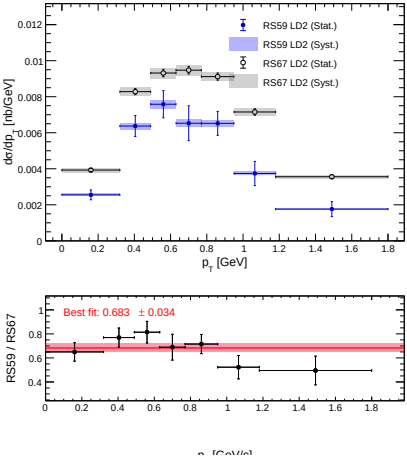


Cross-Section Ratio RS59 to RS67

LH2

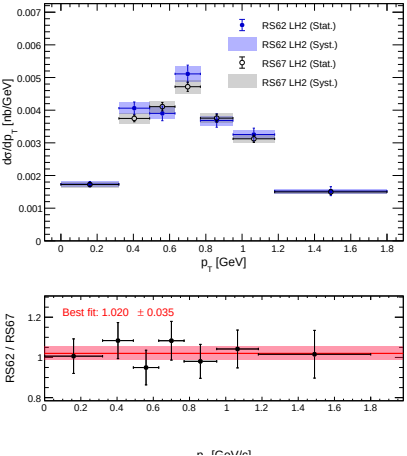


LD2

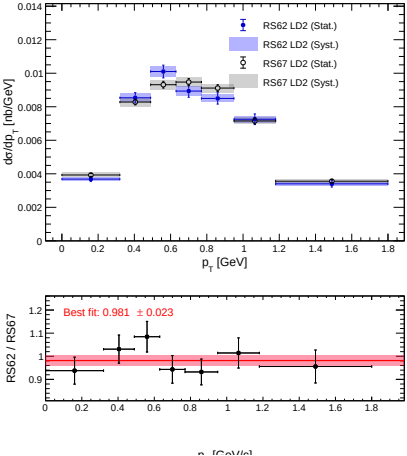


Cross-Section Ratio RS62 to RS67

LH2

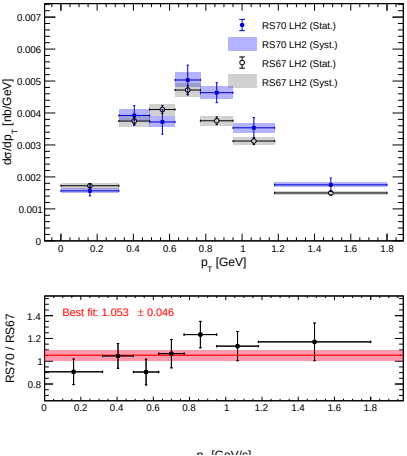


LD2

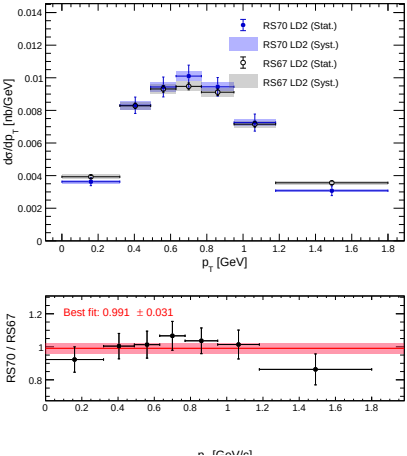


Cross-Section Ratio RS70 to RS67

LH2

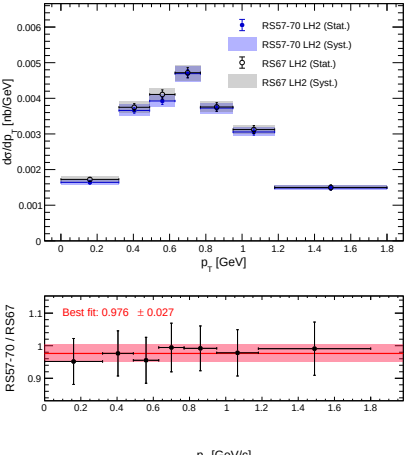


LD2

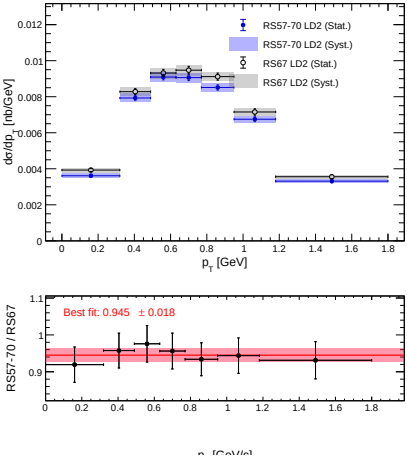


Cross-Section Ratio RS57-70 to RS67

LH2



LD2



Cross-Section comparison to RS67

Table 1: Best-fit single differential cross-section ratios vs. pT (RS/RS67) for LH_2 and LD_2 targets.

Roadset Ratio	**LH_2 Best Fit ($\pm$ Error)**	**LD_2 Best Fit ($\pm$ Error)**
RS57 / RS67	0.773 ± 0.036	0.736 ± 0.022
RS59 / RS67	0.748 ± 0.049	0.683 ± 0.034
RS62 / RS67	1.020 ± 0.035	0.981 ± 0.023
RS70 / RS67	1.053 ± 0.046	0.991 ± 0.031
RS57-70 / RS67	0.976 ± 0.027	0.945 ± 0.018

Roadset	**57**	**59**	**62**	**70**	**57-70**
Cross-Section Ratio: Roadset/RS67 (LH2)	0.754 ± 0.031	0.729 ± 0.047	0.994 ± 0.028	1.035 ± 0.040	0.956 ± 0.020
Cross-Section Ratio: Roadset/RS67 (LD2)	0.731 ± 0.020	0.677 ± 0.033	0.973 ± 0.021	0.986 ± 0.029	0.946 ± 0.016

Figure 1: Previous results presented in DocDB 11547

Conclusions

- After re-calculating reconstruction efficiency and acceptance corrections, there is an improvement in agreement between cross-section results from different roadsets.
- RS57, RS59 and RS62, RS67, RS70 results agrees each other but, there are differences between these two groups of roadsets.
- We still need to understand the discrepancy between the cross-section results between runs 2 and 3.

Next Steps

- Calculating reweights to minimize the observed differences between subtracted data and MC events presented in DocDB 11586.
- Re-calculating reconstruction efficiency and acceptance corrections after applying new reweights to MC events.
- Integrating unfolding procedure to correct for bin migration effects (work on progress).